TEMASEK JUNIOR COLLEGE

Preliminary Examinations 2013

BIOLOGY **9648/01**

Higher 2 **Thursday, 26 September**

Paper 1 Multiple Choice **1 hour 15 minutes**

Additional Materials: **Optical Mark Sheet (OMS)**

READ THESE INSTRUCTIONS FIRST

Write in a soft pencil.
Do not use staples, paper clips, highlighters, glue or correction fluid.
Write your name, Centre number and index number on the Answer Sheet in the spaces provided.

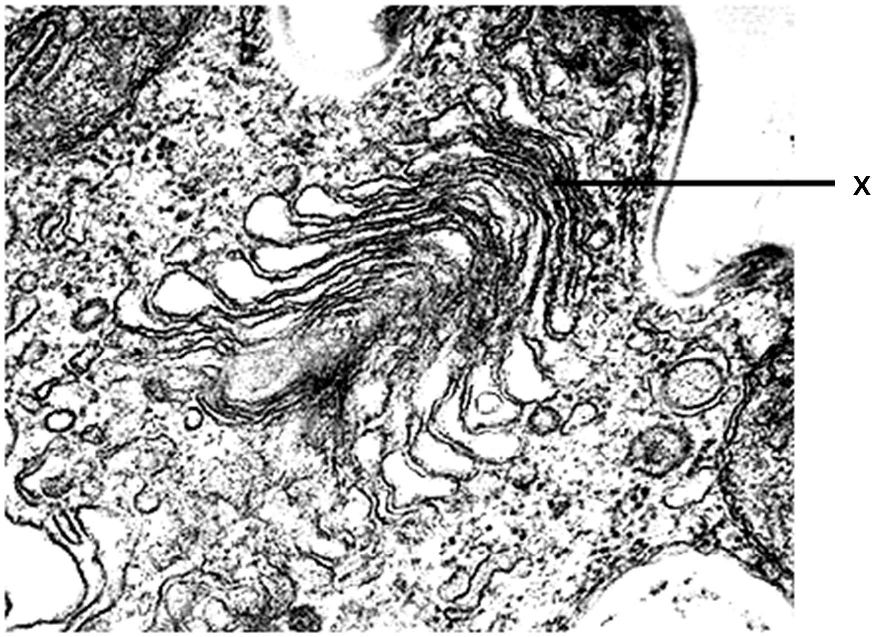
There are **forty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.
Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for the wrong answer.
Any rough working should be done in this booklet.
Calculators may be used.

This document consists of **23** printed pages and **1** blank page.

1 Which of the following are the most likely consequences for a cell lacking structure **X**?



1 The cell dies because it is unable to make glycoproteins to detect stimuli from its environment.
2 The cell dies from a lack of enzymes to digest food taken in by endocytosis.
3 The cell dies from the accumulation of worn out organelles within itself.
4 The cell is unable to reproduce itself.
5 The cell is unable to export its enzymes or peptide hormones.

A 2 and 3

B 2 and 5

C 3, 4 and 5

D 1, 2, 3 and 5

2 A microscope is set up to examine the surface of a leaf. The 40x objective lens and a 7x eyepiece is being used. The diameter of the field of view is 0.4mm.

What is the approximate density of the stomata if 15 of them are visible under these conditions?

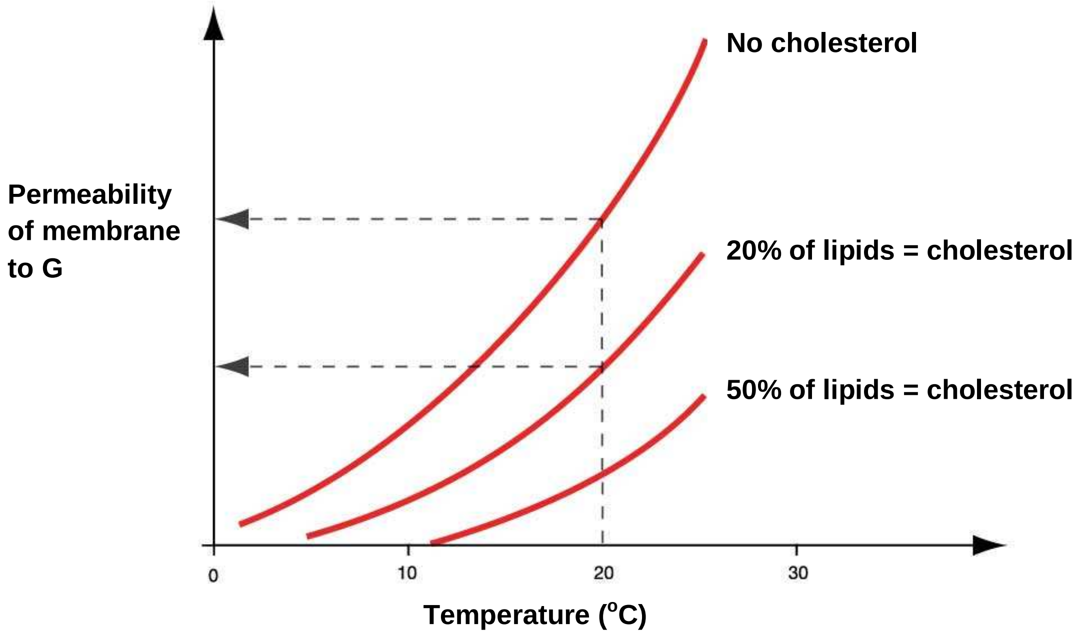
A 33 stomata/mm^2

B 120 stomata/mm^2

C 528 stomata/mm^2

D 1680 stomata/mm^2

3 The graph below shows the permeability of three different membranes to chemical **G** at different temperatures. These three membranes differ in the amount of cholesterol present in the phospholipid bilayers.



Which of the following is a possible explanation for the observed data?

A Increase in temperature increases the permeability of the membrane to **G** as cholesterol increases the fluidity of the phospholipids.

B At 20°C, an increase in the proportion of cholesterol in the membrane increases the permeability of membrane to **G** as both cholesterol and **G** are non-polar.

C Increase in the proportion of cholesterol decreases the permeability of the membrane to **G** as cholesterol decreases the fluidity of the phospholipids.

D With an increase in the proportion of cholesterol in the membrane, a lower temperature is required to achieve the same level of permeability for **G** as **G** will gain a higher kinetic energy to penetrate the membrane.

4 Which of the following are true of triglycerides and phospholipids?

1 Both contain glycerol.
2 Both are amphipathic.
3 Triglycerides have higher carbon content than phospholipids for storage purposes.
4 Phospholipids have hydrophilic regions to interact with cell cytoplasm unlike triglycerides.
5 Triglycerides are formed from glycerol and fatty acids but phospholipids are formed from glycerol and phosphoric acid.

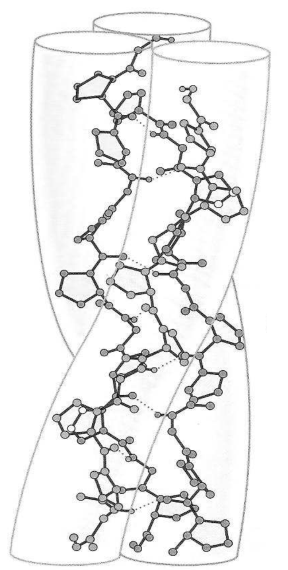
A 1, 3 and 5

B 1, 2 and 3

C 1, 3, and 4

D 2, ,3 and 4

5 The molecule shown below is found in various parts of the human body, and is helical in structure.



Which of the following statement about this molecule is true?

A Complementary base pairing occurs through the formation of hydrogen bonds between chains.

B Hydrogen bonds within each chain give each chain its helical structure.

C Glutamine occurs very frequently in the sequence of the subunits in the molecule.

D The entire molecule interacts with proteins to form complexes necessary for the control of cell activities.

6 The enzyme amylase hydrolyses starch into disaccharides. Amylase is unable to break down cellulose because

A cellulose does not contain glycosidic bonds.

B starch consists of glucose; cellulose consists of other monosaccharides.

C the bonds between sugars in cellulose are much stronger.

D the sugars in cellulose are bonded together differently than in starch, hence cellulose have a different shape.

7 The figure below shows the various ions, molecules and transport proteins that are present in a cholinergic synapse.

Which of the following statements correctly describe the transport of substances across synaptic membranes?

1 Some substances may diffuse readily across the membrane

2 Ions move across the membrane through protein channels via facilitated diffusion.

3 Some ions are actively transported across the membrane by carrier proteins.

4 There are at least two transport mechanisms that require energy expenditure.

A 2 and 3 only

B 1, 2 and 3 only

C 1, 3 and 4 only

D All of the above

8 A pear was cut into six equal segments, two of which were blanched by immersing them in boiling water for one minute. Each segment of the pear was then cut into pieces and these were treated in several ways as indicated in the table below. The time taken for each piece of pear to develop a standard degree of browning was then recorded. You may assume that all pieces had the same initial colour.

Liquid in which pear pieces were immersed	**Initial treatment of pear segment**	**Time for appearance of browning at the temperature indicated (min)**		
		20˚C	40˚C	100˚C
Water	Unblanched	25	9	X
	Blanched	X	X	X
0.1M nitric acid	Unblanched	X	X	X
2% dithionite solution	Unblanched	X	135	X

X indicates that the piece of pear did not turn brown within 180 minutes.

The enzyme involved in the browning of the pear pieces is

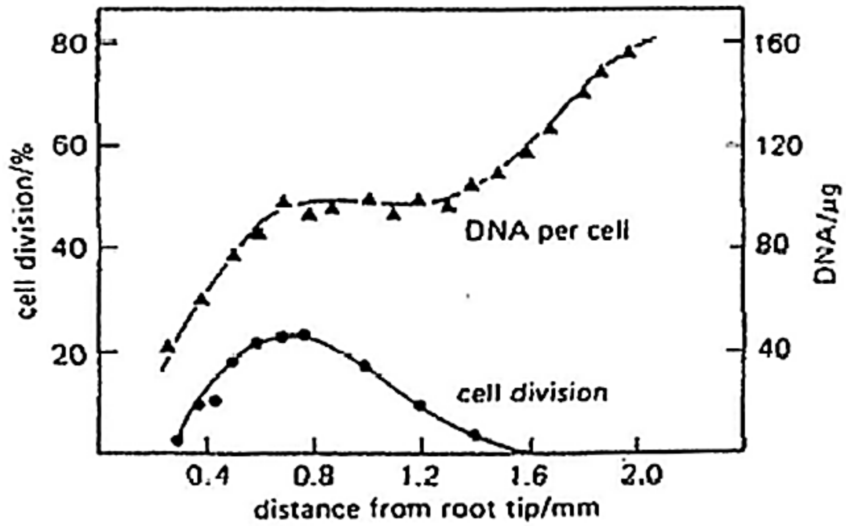
1 thermostable.
2 destroyed by blanching.
3 inhibited more by 2% dithionite solution than by 0.1M nitric acid.
4 is not the only agent responsible for the browning of the pear in the experiment.

A 1 and 3 only
B 2 and 3 only
C 2 and 4 only
D 2, 3 and 4 only

9 The second division of meiosis differs from mitosis because in meiosis

A chiasmata form between the chromatids.
B each chromosome replicates at metaphase.
C individual chromosomes line up at random on the equator.
D the separating chromatids differ genetically.

10 The graph below shows the changes in DNA content and rate of division in cells at varying distances from the tip of an onion root.



Which one of the following can be determined from the graph?

A Cells nearest the tip are dividing by meiosis.

B DNA quantity varies inversely with the cell division rate.

C DNA synthesis precedes cell division.

D The cells become polyploidy.

11 The table below shows a list of characteristics displayed by mutant strains of *E.coli* during DNA replication and the possible reasons.

No	Characteristics	Enzymes or functions affected by mutation
1	Okazaki fragments accumulate and DNA synthesis is never completed	DNA ligase activity is missing
2	Supercoils are found to remain at the flanks of the replication bubbles	DNA helicase is hyperactive
3	Synthesis of new strand is very slow	Substrate binding site of RNA polymerase slightly altered
4	No initiation of replication occurs	A-T rich region at origin of replication deleted

Which of the reasons correctly explain the characteristics displayed by the mutant *E. coli* strains?

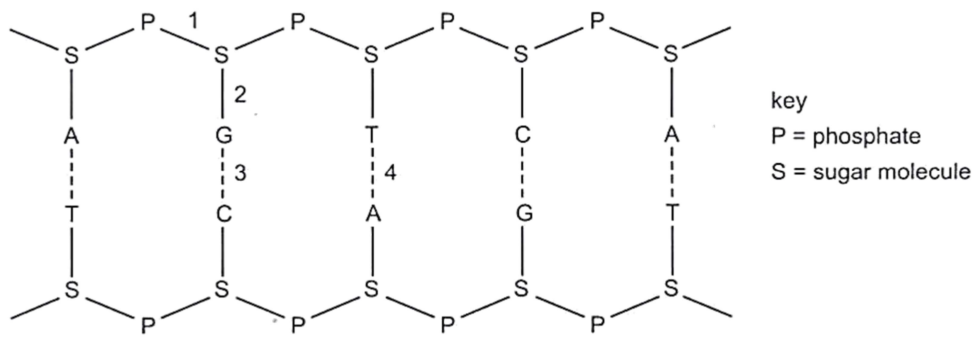
A 1 and 2

B 1 and 3

C 1, 3, and 4

D All of the above

12 The diagram shows part of a nucleic acid.



Which row correctly describes the bonds shown in the diagram at positions 1, 2, 3 and 4?

	Is formed by condensation	**Forms a di-ester**	**Occurs during transcription**	**Involves attraction between polar molecules**
A	1	2	1, 2 and 3	3
B	1 and 2	1	3 and 4	3 and 4
C	2, 3 and 4	1	3 and 4	1, 3 and 4
D	2	3	1, 3 and 4	4

13 In three different genetic dictionaries, the genetic code for the amino acid cysteine is given as:

I **ACA** or **ACG**
II **TGT** or **TGC**
III **UGU** or **UGC**

The explanation for this may be:

1 Some genetic dictionaries show mRNA codons, others show DNA triplets.

2 The genetic code can be read in either the 3’ or 5’ direction along the DNA.

3 Some genetic dictionaries show the triplet code on the DNA strand complementary to the mRNA code, others show the triplet code on the other DNA strand.

4 The genetic code is a degenerate triplet code.

A 3 only

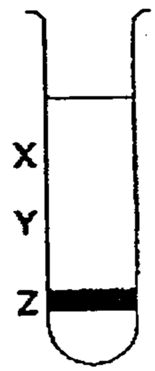
B 2 and 4 only

C 1, 2 and 3 only

D 1, 3 and 4 only

14 A culture of bacteria had all its DNA labelled with the heavy isotope of nitrogen, ^{15}N. The bacterial culture was then allowed to reproduce using nucleotides containing normal ^{14}N. The DNA was examined using a centrifuge after one generation and again after two generations.

The diagram shows the position of the DNA band at **Z** in the centrifuge tube when the DNA was first labelled.



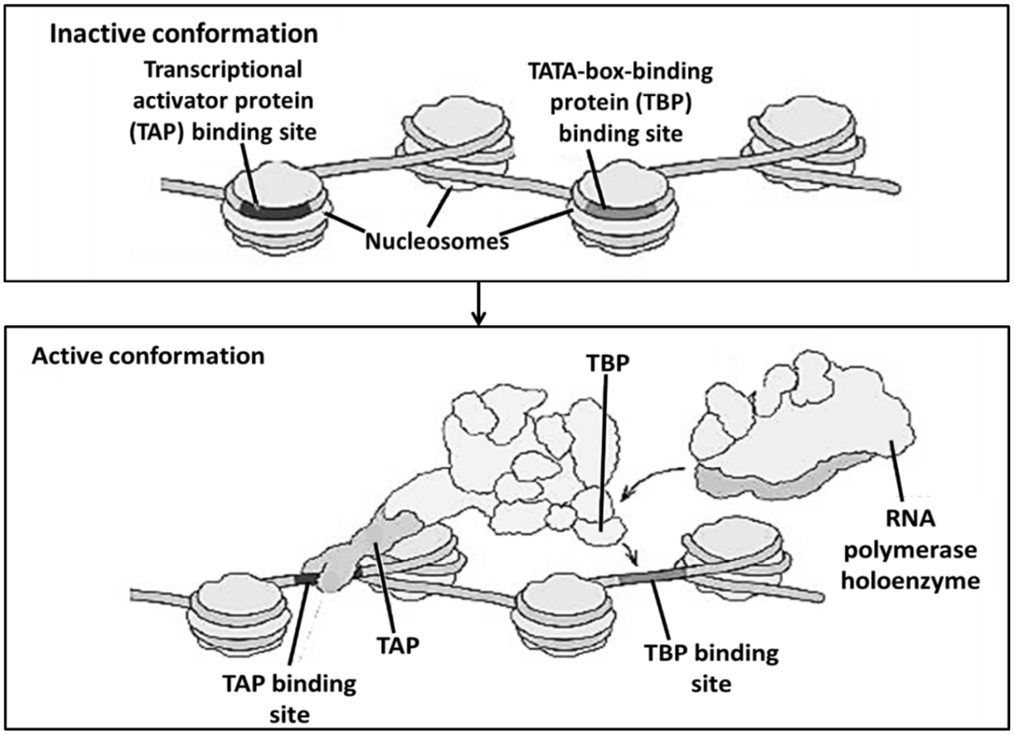
Where would the DNA be found after the second and after the third cell generations?

	After second generation	After third generation
A	Half at **X** and half at **Y**	One quarter at **X** and three quarter at **Y**
B	Half at **X** and half at **Y**	One quarter at **Y** and three quarter at **X**
C	One quarter at **X** and three quarter at **Y**	Half at **X** and half at **Y**
D	One quarter at **Y** and three quarter at **X**	Half at **X** and half at **Y**

15 Which pair of statements correctly describes centromeres and telomeres?

	Centromeres	Telomeres
A	Only present when DNA condense to become sister chromatids	Only present when chromosome unwinds during interphase
B	A change in the centromeric sequence may result in aneuploidy	A change in the telomeric sequence may result in chromosomal mutation
C	Are split during cell nuclear division	Are shortened during interphase
D	Always found in the middle of a chromosome	Always found at the ends of a chromosome

16 The figure shows the process of chromatin modification prior to transcription.



Which of the following statements regarding chromatin modification is correct?

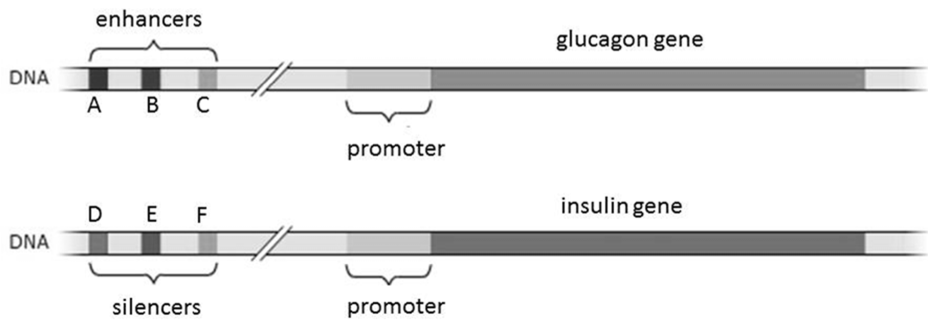
A Chromatin modification complexes result in the reversible acetylation or deacetylation of DNA.

B TAP binding results in the immediate downregulation of transcription.

C The binding of TAP and TBP to their binding sites is dependent on the recognition of histone tails.

D In the active conformation of chromatin, the repositioning of nucleosomes allows for the exposure of TAP and TBP binding sites, leading to protein binding.

17 The diagram below shows the control elements and two genes found in the human genome.



Which of the following statements about the above genes are true?

1 The glucagon gene is found only in the α-cells of the Islets of Langerhans while the insulin gene is found only in the β-cells of the Islets of Langerhans.

2 Binding specific transcription factors and RNA polymerase at the enhancer initiates transcription of glucagon.

3 The glucagon gene will be transcribed at a high level when specific transcription factors bind to control elements **A**, **B**, and **C**.

4 The expression of insulin can be suppressed when repressors bind to control elements **D**, **E** and **F**.

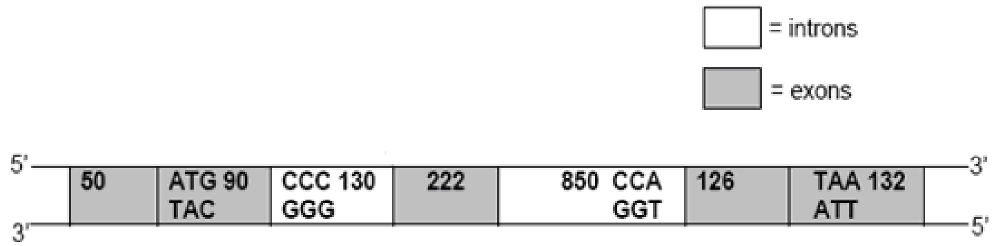
A 2 and 3 only

B 3 and 4 only

C 1, 3 and 4 only

D 1, 2 and 4 only

18 The diagram below shows the genomic structure of a human gene. The numbers within the boxes indicate the length of nucleotides of each region, inclusive of the bases stated in the diagram. The DNA sequences corresponding to the start codon and the stop codon are also indicated.



What is the length (in nucleotides) of the primary RNA transcript (pre-mRNA) and how many amino acids are present in the protein?

	Length of primary RNA transcript	**Number of amino acid in protein**
A	1 600	146
B	620	146
C	1 600	206
D	620	206

19 Which of the following statements(s) is / are correct about prokaryotic gene expression?

1 Prokaryotic mRNAs are often translated before transcription is complete.
2 Prokaryotic mRNAs contain the transcript of only one gene.
3 Prokaryotic mRNAs must have exons spliced together.
4 Prokaryotic mRNAs have processed 5' caps.

A 1 only
B 1 and 2 only
C 3 and 4 only
D All of the above

20 In *E.coli* bacteria, regulation of gene expression involving the catabolite activator protein (CAP), is actually a type of positive regulation because

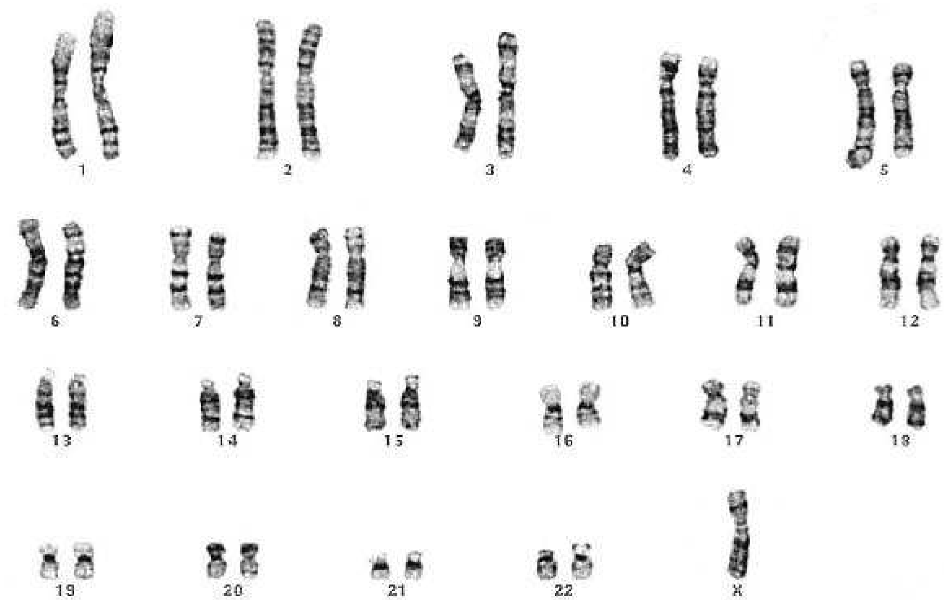
A cAMP-CAP helps RNA polymerase initiate transcription.

B glucose stimulates the production of cAMP.

C CAP increases the production of cAMP.

D glucose binds to CAP and activates it.

21 In Turner's syndrome, a mutation results in the loss of an X chromosome in females. This can lead to sterility and growth abnormalities.



Which of the following correctly describes the type of mutation observed in the above karyotype?

A Chromosomal mutation, polyploidy

B Chromosomal mutation, aneuploidy

C Gene mutation, polyploidy

D Gene mutation, aneuploidy

22 Which of the following is **not** a possible effect of a mutated *p53* gene?

A p53 protein is unable to bind to specific DNA sequence.

B p53 protein subunits are unable to interact with each other to form the complete functional protein.

C Products of viral oncogenes bind to and inactivate the p53 protein in the cytoplasm, resulting in degradation of p53 protein.

D p53 protein is unable to fold into its specific 3D conformation.

23 Which of the following contributes towards cancer progression?

1 A mutation in the gene that slows down cell cycle

2 Faulty DNA repair

3 Loss of control over telemore length

4 Expression of gene that results in growth of new blood vessels

A 2 and 3 only

B 1 and 4 only

C 1, 2 and 3 only

D All of the above

24 Cells of an *E. coli* strain that are *trp*- *lac* Z- *met*+ were mixed with cells of an *E. coli* strain that are *trp*+ *lac* Z+ *met*- and cultured for several hours. Then cells were removed, washed, and transferred to minimal media containing lactose as the only sugar source. A few cells were able to grow on minimal media with lactose, and formed colonies.

Which process/processes might have taken place?

1 transformation

2 transduction

3 mutation

4 conjugation

A 1 only

B 2 only

C 1 and 3

D 1 and 4

25 RNA viruses appear to have higher rates of mutation because

A RNA viruses replicate faster.

B RNA viruses respond more to mutagens.

C RNA nucleotides are more unstable than DNA nucleotides.

D replication of their nucleic acid does not involve the proofreading steps of DNA replication.

26 The influenza virus is an enveloped virus with an RNA nucleocapsid. It enters a host cell by endocytosis and new viruses emerge by budding.

Some events that take place during the reproductive cycle are listed.

1 The virus enters the cytoplasm surrounded by a coated vesicle.

2 The nucleocapsid is released and viral RNA replicated by RNA polymerase.

3 Each replicated viral RNA is given a new envelope.

4 Viral hemagglutinin proteins bind to receptors on the host cell surface membrane.

5 Hemagglutinin and neuraminidase are produced by the host cell's ribosomes.

6 The replicated viral RNA is transcribed to mRNA.

Which sequence of events correctly describes the reproductive cycle of the virus?

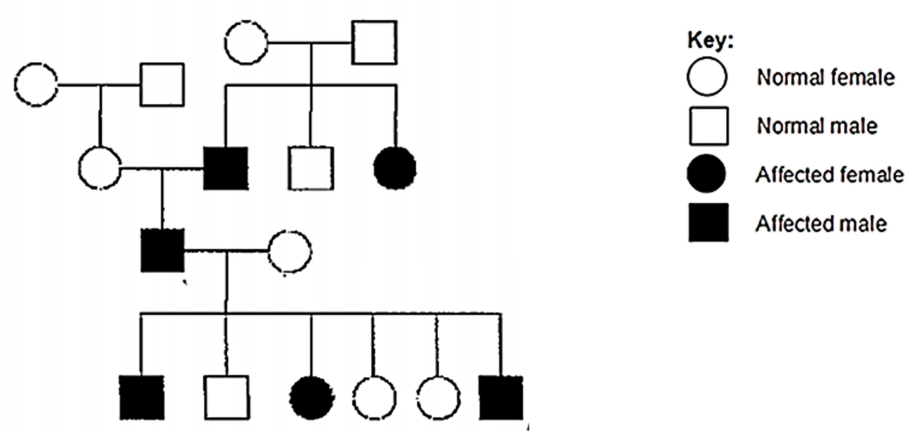
A $1 \rightarrow 2 \rightarrow 3 \rightarrow 6 \rightarrow 5 \rightarrow 4$

B $1 \rightarrow 2 \rightarrow 6 \rightarrow 5 \rightarrow 4 \rightarrow 3$

C $4 \rightarrow 2 \rightarrow 1 \rightarrow 5 \rightarrow 6 \rightarrow 3$

D $4 \rightarrow 1 \rightarrow 2 \rightarrow 6 \rightarrow 5 \rightarrow 3$

27 Based on the pedigree chart below, what is the most likely inheritance pattern?



A Autosomal dominant

B Autosomal recessive

C X-linked dominant

D X-linked recessive

28 The genotype of F_1 individuals is **AaBbCcDd**. Assuming independent assortment of these four genes, what is the probability that the F_2 offspring would express the phenotype of genotype **aabbccdd**?

A 1/4

B 1/16

C 1/256

D 1/512

29 The phenotypes of 200 offspring of a dihybrid test cross were recorded. The cross involved petal colour and fertility of the anthers of sweat pea flowers. The table shows the observed and expected numbers of each phenotype.

Phenotype	**Purple petals fertile anthers**	**Purple petals sterile anthers**	**Maroon petals fertile anthers**	**Maroon petals sterile anthers**
Observed	87	14	16	83
Expected	50	50	50	50

A chi-squared (χ^2) test was performed and the *p value* was found to be <0.001.
What conclusions may be drawn from this probability?

1 The difference is significant.
2 The difference is insignificant.
3 The difference is due to chance.
4 The difference is not due to chance.
5 The genes for petal colour and fertility of anthers are likely to be linked.
6 The genes for petal colour and fertility of anthers are likely to be unlinked.

A 1, 3 and 6
B 1, 4 and 5
C 2, 3 and 6
D 2, 4 and 5

30 Molecular homology of haemoglobin is used to determine evolutionary relatedness in a range of mammals. Either nucleic acids or proteins can be sequenced. The sequence of nucleic acids and proteins are compared.

Which sequence will vary **least** in related mammals?

A Amino acid

B DNA

C Edited mRNA transcript

D Primary mRNA transcript

31 One similarity and one difference between classification and phylogeny are that

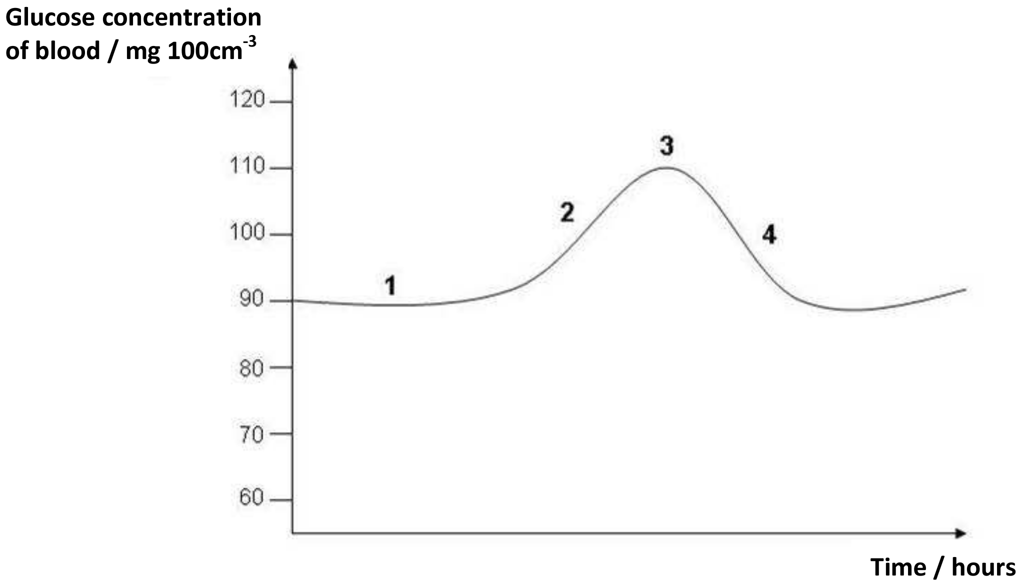
A both organise organisms into groups based on their characteristics but phylogeny takes into consideration the evolutionary relationships between groups.

B both put organisms together according to increasingly inclusive groups but classification does not show a branching pattern of evolutionary relationships.

C both rely on physical characteristics of organisms but phylogeny does not require scientific names.

D both take into account the common ancestors of organisms but classification is hierachical in nature.

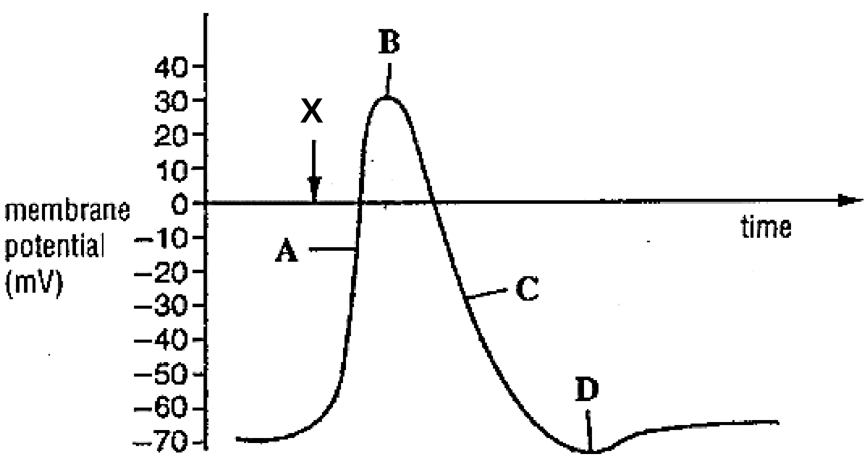
32 The graph below shows the changes of blood glucose level in a normal healthy person who consumes a meal.



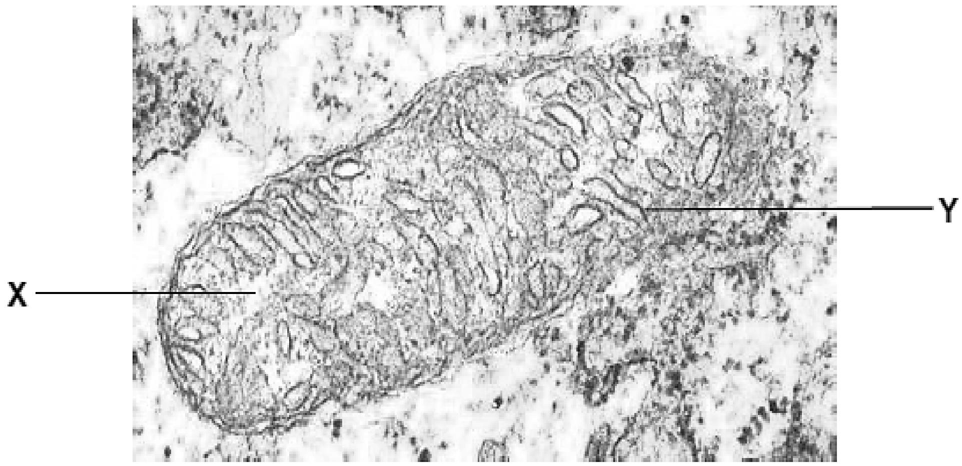
What is represented at each stage?

	Negative feedback	Reference point	Effector cells activated
A	4	1	3
B	4	1	2
C	3	1	4
D	3	1	2

33 The graph shows the changes in membrane permeability when a stimulus **X** is applied to a neurone. Where is the point of maximum permeability to sodium ions?



34 The electron micrograph shows a mitochondrion with structures **X** and **Y** labelled.

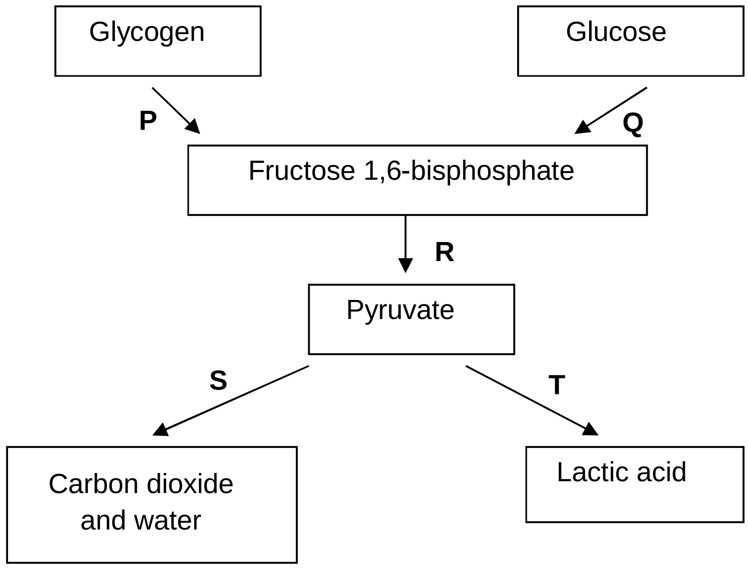


The phosphate : oxygen (P/O) ratio represents the number of ATP produced by oxidative phosphorylation per molecule of oxygen consumed. You may assume that the re-oxidation of each reduced coenzyme requires one oxygen molecule.

Under aerobic conditions, which of the following correctly represents the P/O ratio in structures **X** and **Y**?

	P/O ratio obtained by the complete oxidation of 1 molecule of glucose	
	Structure X	**Structure Y**
A	2.8	2.8
B	2.2	2.8
C	2.8	3.3
D	3.0	3.3

35 With reference to the diagram below, relate processes **P, Q, R, S, T** to statements **1**, **2** and **3**.



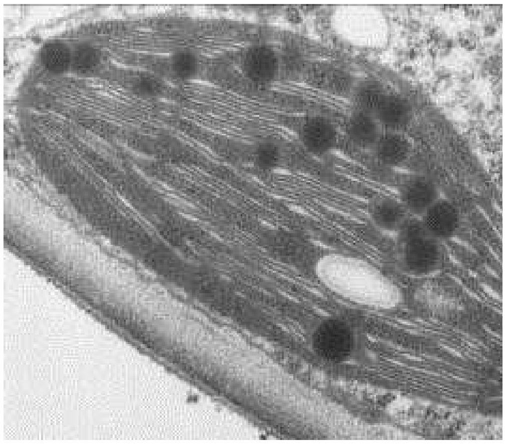
1 NAD^+ is regenerated without the use of the electron transport system.

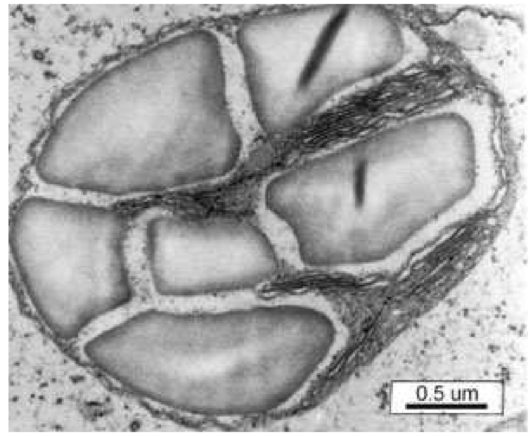
2 ATP is synthesised via substrate level phosphorylation.

3 It can take place under anaerobic conditions.

	1	**2**	**3**
A	R	S	T
B	T	R	P
C	None of the above	S	R
D	S	Q	None of the above

36 The electron micrographs show two types of chloroplasts found in maize plants.


Chloroplast **P**



Chloroplast **Q**

Which of the following can be concluded from these photomicrographs of the maize chloroplasts?

1 There is intense photosynthetic activity in chloroplast **P** as compared to chloroplast **Q**.

2 GP, the final product of photosynthesis in chloroplast **P**, is transported to chloroplast **Q** where is it eventually converted to starch for storage.

3 Chloroplast **P** is found in regions of the maize plant which are exposed to more direct sunlight as compared to chloroplast **Q**.

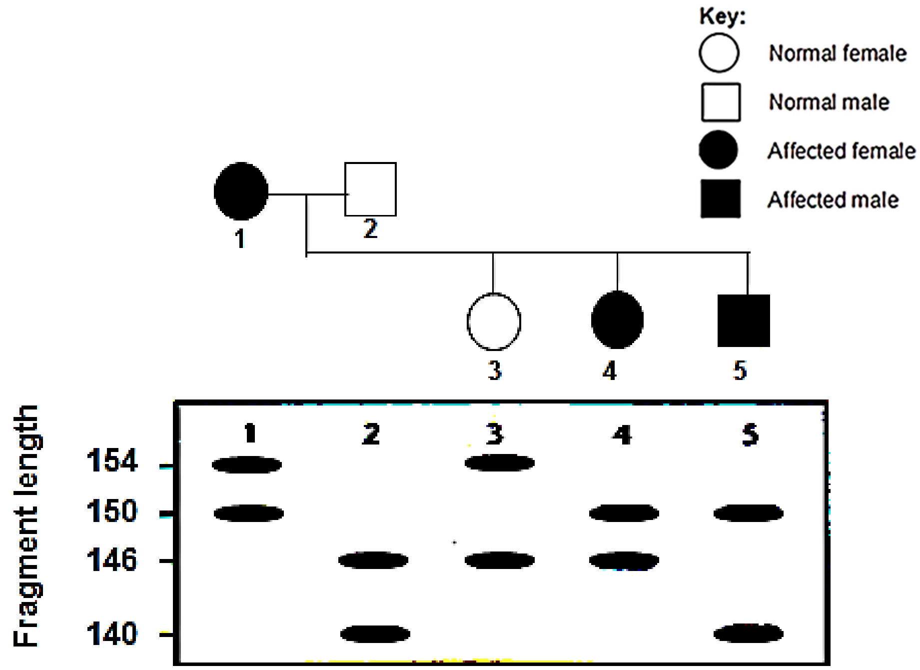
A 2 only

B 1 and 2 only

C 1 and 3 only

D All of the above

37 An RFLP genetic marker located on chromosome 13 is used to trace the inheritance of a genetic disease within an affected family. The following pedigree shows two generations of the family with affected individuals. Results of the Southern Blot analysis for the corresponding family members are shown in the pedigree.



The allelic variants of the polymorphic RFLP locus are designated by the respective fragment lengths produced in this linkage analysis. Which allele of the RFLP locus is linked to the disease allele?

A Allele **140**

B Allele **146**

C Allele **150**

D Allele **154**

38 Which of the following is an argument in favour of using embryonic stem cells over adult stem cells?

A Embryonic stem cells are not living cells.

B Adult stem cells cannot be cultured.

C Adult stem cells reproduce much faster than embryonic stem cells.

D Embryonic stem cells can differentiate into many more types of cells.

39 Which of the following statements is **not** a possible issue of concern over the creation of genetically modified farmed animals?

1 Genetic engineering may result in the creation of new proteins that are harmful to the organisms that produce or consume them.

2 Cross species gene transfer may compromise the genome integrity of the species involved.

3 Over production of certain gene products may cause undue stress to the genetically modified farmed animals.

4 Some genetically modified food products may not be acceptable to certain groups of people.

A 1 and 4

B 2 and 3

C 1, 3 and 4

D None of the above

40 In terms of containment (prevention of genetic pollution of wild plant or non-GM crop populations), which of the following is an advantage of chloroplast transformation (introducing foreign genes into chloroplasts of host plants) over nuclear transformation?

A Chloroplasts are surrounded by a double membrane.

B There are no chloroplasts in pollen of most plant species.

C Chloroplasts are smaller than the nucleus.

D There is no DNA in chloroplasts.

~ End of Paper 1 ~

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